Recommender Systems - Course Project Report

Cumulative report template - Parts 1, 2, and 3

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For Part 3 track: select one	A / B / C
GitHub repository link	https://github.com/usernameis33321/recsys_project

How to use this template

- *This is one cumulative report. Extend and revise the same document at each project stage.*
- *Maximum length: 4 pages at Part 1, 8 pages at Part 2, and 12 pages for the final report, excluding references.*
- *Keep the section headings below. You may add subsections where useful.*
- *All explanations and justifications requested in the project specification belong in this report.*
- *The Results Sheet contains the complete standardized numerical record. In the report, include only the tables and figures needed to support your analysis.*
- *Delete all italic instructional text before the final submission.*

AI Usage

I used ChatGPT as a coding assistant, mainly to help identify and fix errors. The main logic and structure of the code were developed by myself.

I also used ChatGPT to check the grammar of the README.md file and this report.

1. Part 1 - Data, Protocol and Baselines

1.1 Data characterization

Analyze the MovieLens 1M data we could obtain, that the data has a sparse interaction structure for the matrix density is 4.47%, while many uses provide ratings for only few popular movies.

The long-tail distribution of item interactions (figure 1), and the distribution of interactions per user (figure 2) shown below. We could obtain that observe that only a small proportion of popular movies receive many interactions (ratings), and most users have relatively few interactions, whereas a small number of users account for a large proportion of the total interactions.

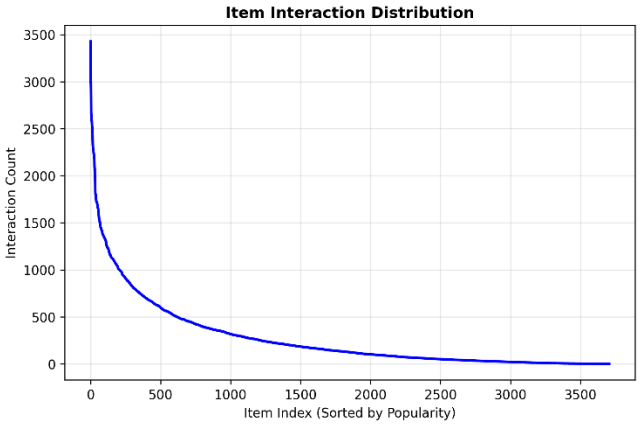


Figure 2 The item interaction distribution follows a clear long-tail pattern. A small number of popular movies receive a large number of ratings, while the majority of movies have relatively few interactions. This indicates a significant difference in popularity among movies.

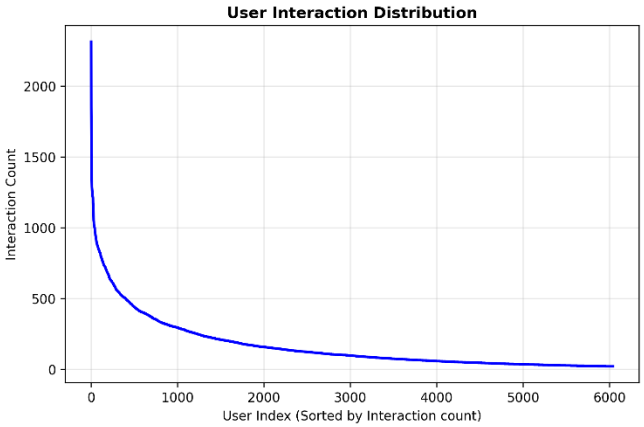


Figure 1. The user interaction distribution also follows a long-tail pattern. Most users have relatively few interactions, while a small proportion of users account for a large number of interactions. This indicates substantial differences in user activity levels.

The data also show that user interactions are time-dependent and movie popularity changes over time. Only 6 movies are shared between the top 10 most popular movies in the first and second halves of the dataset. The interaction counts with time shown in figure 3. Therefore, a chronological split was applied to the dataset, with 70% allocated to training, 15% to validation, and 15% to testing.

The training set covers the period from April 25, 2000, to November 22, 2000, while the validation set covers approximately 18 days, from November 22 to December 10, 2000. The test set spans a much longer period, from December 10, 2000, to February 28, 2003.

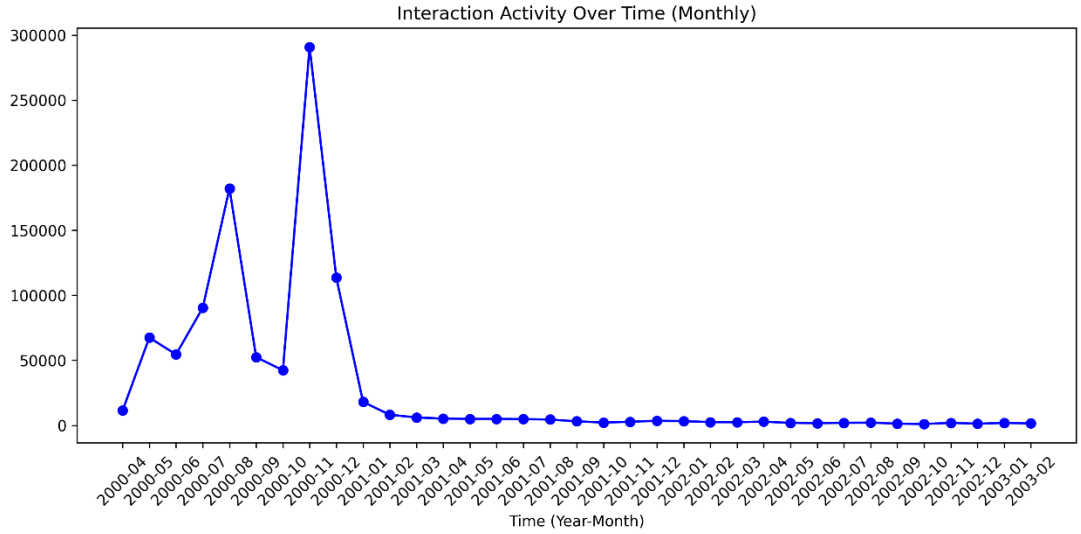


Figure 3. The monthly interaction activity varies significantly over time. It reaches a peak in October 2000, followed by a sharp decline. From early 2001 onwards, the monthly interaction volume remains relatively low and stable.

1.2 Evaluation protocol

The recommendation task was formulated as a Top-10 recommendation problem. For each user, movies already rated in the training set were excluded from the candidate set, and the remaining movies were ranked based on the recommendation scores.

Ratings of 4 or 5 in the evaluation set were considered relevant items, while ratings below 4 were considered non-relevant.

The recommendation performance was evaluated using Precision@10, which refer recommendation accuracy; Recall@10, which refer relevant item coverage; NDCG@10, which refer ranking quality. Additionally, Catalogue Coverage was used to measure the proportion of available movies included in the recommendations.

The same candidate set and evaluation protocol were applied to all baselines.

1.3 Baseline methods and settings

Item-based KNN

Item-based KNN was implemented using item-item similarity to generate personalized recommendations. The neighborhood size was tested with k = [1, 5, 50, 150, 200], and three similarity measures were evaluated: cosine, Euclidean, and correlation. The final configuration was selected based on validation performance. The table of all metrics with different k and similarity measures shown below:

n_neighbors	similarity measure	Precision@10	Recall@10	NDCG@10	Catalogue Coverage
1	cosine	0.0407	0.0164	0.0481	0.1431
1	euclidean	0.0051	0.0015	0.0047	0.2846
1	correlation	0.0343	0.0117	0.0387	0.1621
5	cosine	0.0531	0.0183	0.0579	0.1159
5	euclidean	0.0015	0.0002	0.0015	0.2915
5	correlation	0.0414	0.014	0.0453	0.1495
50	cosine	0.0578	0.0181	0.0633	0.0735
50	euclidean	0.0009	0.0006	0.0011	0.2706
50	correlation	0.0404	0.0143	0.0444	0.1497
150	cosine	0.06	0.0182	0.0659	0.0606
150	euclidean	0.0008	0.0002	0.0008	0.2956
150	correlation	0.0406	0.0131	0.0445	0.1423
200	cosine	0.0603	0.0184	0.0659	0.0617
200	euclidean	0.0011	0.0002	0.0012	0.2976
200	correlation	0.0416	0.0135	0.0452	0.1376

The results show that cosine similarity achieved the best recommendation. Its best Precision@10 was 0.0603, 0.0184 for Recall@10, Catalogue Coverage was 0.0606 and CDCG@10 was 0.0659.

For cosine similarity, increasing the number of neighbors generally improved Precision@10 and NDCG@10. However, the improvements became very small when the number of neighbors increased from 150 (show in light gray) to 200 (show in darker gray). At the same time, Catalogue Coverage decreased as the number of neighbors increased.

This indicates a trade-off between recommendation accuracy and catalogue coverage. Based on these results, cosine similarity with 150 neighbors was selected as a reasonable configuration.

Most Popular

The differences between the two variants were small, indicating that filtering out low ratings did not substantially improve recommendation performance.

Both variants achieved low catalogue coverage (3.85% and 3.63%, respectively). This indicates that the Most Popular baseline tends to recommend a relatively small set of popular movies, limiting the variety of recommended items.

Popularity variant	Precision@10	Recall@10	NDCG@10	Catalogue Coverage
all_ratings	0.2753	0.0608	0.2957	0.0385
positive_ratings	0.2781	0.0618	0.2982	0.0363

1.4 Results and analysis

The final evaluation shows that the Most Popular baseline outperformed Item-based KNN in all three ranking metrics. Most Popular achieved a Precision@10 of 0.2199, Recall@10 of 0.0518, and NDCG@10 of 0.2332. In comparison, Item-based KNN achieved 0.1113, 0.0309, and 0.1234, respectively.

However, Item-based KNN achieved higher catalogue coverage (0.0718) than Most Popular (0.0333). This indicates that KNN recommended a wider range of movies, whereas Most Popular concentrated its recommendations on a smaller group of popular items.

Overall, the results reveal a trade-off between recommendation accuracy and catalogue coverage. Although Item-based KNN provides broader item coverage, the simple Most Popular baseline produces more accurate recommendations in this experiment.

Model	Precision@10	Recall@10	NDCG@10	Catalogue Coverage
"Item-based KNN (Cosine & k=150)"	0.1113	0.0309	0.1234	0.0718
Most Popular (Positive ratings)	0.2199	0.0518	0.2332	0.0333

2. Part 2 - Models and Honest Comparison

2.1 Models

Describe the additional models and paradigms. Clearly identify the self-implemented algorithm and explain its implementation.

2.2 Hyperparameter tuning and variability

Describe the validation-only tuning procedure, search spaces/budgets, random seeds, and confidence intervals or statistical testing.

2.3 Overall results

Present and discuss the main comparison. Explain important differences and avoid unsupported claims.

2.4 Disaggregated analysis

Report and interpret results broken down by at least two of the required dimensions: items with few vs. many training interactions, low- vs. high-activity users, popular vs. long-tail items, and early vs. late test period. Define any thresholds used and focus on findings that change or qualify the overall conclusions.

2.5 Sparsity ablation

Report and discuss the 100%, 50%, and 20% training-interaction experiments, including changes in the ranking between models.

3. Part 3 - Beyond Accuracy

3.1 Selected track and motivation

State Track A, B, or C and briefly motivate the selected intervention/extension.

3.2 Method

Describe only your selected track. The Track A/B/C guidance below is provided as instructions; delete the guidance for the two tracks you did not select before submission.

If Track A - Diversity and popularity bias

State the base recommender, selected re-ranking method, genre-vector item representation, trade-off parameter, and exact item-distance measure. Describe how the parameter was swept.

If Track B - Fairness

Define the release-decade provider proxy, report and justify the final decade boundaries and group sizes, state and justify the adopted fairness notion, describe the consumer groups based on MovieLens user attributes, and identify the stage and form of the mitigation.

If Track C - Group recommendations

State the selected synthetic group type and justify the chosen group size (3-5). Describe the three aggregation strategies. Identify which strategy was extended to the sequential scenario and explain how satisfaction is carried forward between recommendation rounds.

3.3 Results and trade-offs

Present the required track-specific results. Track A: report intra-list diversity, catalogue coverage, Gini and calibration error across the parameter sweep, and discuss the accuracy-diversity frontier. Track B: report position-weighted exposure against catalogue share, per-group consumer-side accuracy, and the fairness-performance frontier. Track C: compare group satisfaction, disagreement and minimum member satisfaction across the three aggregation strategies, then analyse how member and group satisfaction develop in the sequential scenario.

3.4 Cost and interpretation

State what is gained, what is lost, and who bears the cost where applicable. Explain what operating choice or strategy you would select and why.

3.5 "One number we cannot explain"

Identify one result you do not fully understand and describe how you would investigate it.

4. Discussion and Conclusions

Summarize the main findings across the project. Discuss limitations, failed approaches where relevant, and what the experiments support.

5. Contribution Statement

For each team member, provide two or three sentences describing their contribution.

6. Use of AI Assistants

State which AI tools were used and for what. If none were used, state that explicitly.

References

List references used in the report. References do not count toward the page limit.